False Consensus: Why Consensus-Based Early Exit Fails in Reasoning LLMs, and a Signal That Works

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Abstract

Reasoning language models produce long chains of thought, making *early exit*—stopping generation once the answer is settled—an attractive way to cut inference cost. A popular family of methods (e.g., Dynasor / Certaindex) periodically probes the partial trajectory for its current answer and stops once probes *agree*, betting that intermediate consensus signals termination. We show this bet is unsafe, and that the failure is not repaired by tuning the stopping rule within this family. On MATH500 we document *false consensus*: even a unanimous window of recent probes is correct only 93.5% of the time, and a naive consensus stop loses 16.4 percentage points (pp) of accuracy ($85.6\% \to 69.2\%$) because continued reasoning frequently overturns the intermediate answer. We then run a *preregistered* search over a seven-dimensional consensus-rule schema (17,712 rules) on frozen reasoning trajectories from two models and three benchmarks, with problem-grouped splits and stopping gates fixed in advance. On the development set, no rule clears the safety bar (the safest stop still costs 1.85 pp), and a faithful Certaindex reproduction on the same trajectories collapses by 56–70 pp. *The failure is specific to the consensus signal, not to early exit.* A different signal—model confidence at reasoning-boundary tokens, with a retained verification branch (in the spirit of DEER)—does far better: on the same models it stays within ~ 1 pp of full accuracy while saving tokens, and an online deployment saves $\sim 44\%$ of tokens at essentially neutral accuracy ($+2.8$ pp on one model). We separate an intrinsic *accuracy tax* from a *probe tax* to explain why consensus stalls, and conclude that safe early exit for reasoning needs a signal about the trajectory's future, not the recent past's agreement. (The boundary-confidence results are exploratory, single-seed, and outside the preregistered sweep.)

1 Introduction

Reasoning language models such as DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B (DeepSeek-AI, 2025; Qwen Team, 2025; OpenAI, 2024) solve hard problems by emitting long chains of thought, often thousands of tokens per query. Much of this computation is spent *after* the model has effectively decided on an answer, which makes *early exit*—halting generation once the answer is settled—an appealing lever for reducing inference cost (Chen et al., 2024; Fu et al., 2024).[1]

A widely used family of early-exit methods works by *probing*. At regular intervals during generation, the partial trajectory is queried for its current answer (e.g., by appending a short "`Final Answer:`" suffix), and the decoder is halted once these probes *agree*—the same answer appears in several consecutive probes, or a share threshold over a recent window is exceeded (Fu et al., 2024). The implicit assumption is that *intermediate consensus is a reliable signal of termination*: once the model repeatedly says the answer is x, further reasoning is unlikely to change it.

This paper asks whether that assumption holds, and whether any probe-based stopping rule can exploit it *safely*. We answer both in the negative for the family of rules we search, and we do so under a preregistered protocol designed to make the negative result credible rather than an artifact of a poorly chosen heuristic.

Finding 1: False consensus. On MATH500 with DeepSeek-R1-Distill-Qwen-7B, we log a dense probe stream for every problem without ever intervening (Section 3). Perfect whole-trajectory consensus is nearly trustworthy (98.9% correct),

[1] We use "false consensus" for the phenomenon that intermediate probe agreement is non-terminal; the term is unrelated to the social-psychology "false consensus effect" (Ross et al., 1977). We also mean answer-level early exit (stopping a chain of thought), not token- or layer-level early exit (Graves, 2016; Schuster et al., 2022).

but the signal a stopping rule actually sees online—agreement over a recent window—is not: a unanimous last-five-probe window is correct only 93.5% of the time, with 6.5% *false consensus*. A naive stop that halts at the first three consecutive agreeing, non-empty probes fires on $416/500$ problems and commits to an answer that is correct only 69.2% of the time, whereas letting those same problems run to completion reaches 85.6%—a 16.4 pp loss. (This is a simplified simulation, not the Certaindex algorithm, which we reproduce faithfully in Section 7.) The mechanism is recovery: of problems whose first probe is wrong, 76.3% eventually flip to correct. Intermediate consensus is simply not terminal.

Finding 2: No safe-and-saving rule in the searched space (preregistered). A natural response is that a smarter rule—higher maturity thresholds, persistence requirements, certainty filters, adaptive probing—would recover safety while keeping the savings. We test this exhaustively over a large preregistered space (Sections 4–5). We freeze one reasoning trajectory per problem, re-probe it offline so that no stopping schedule can alter the underlying reasoning, and sweep a seven-dimensional rule schema of 17,712 candidate rules across two models and three benchmarks. Splits are problem-grouped ($60/20/20$ train/dev/test) and the operating-point gates are fixed in advance; the test split and two confirmation models are never visible to selection. *On the development set, no rule clears the preregistered conservative gate.* The safest rule in the entire space still costs 1.85 pp per model; separately, under dense-probe accounting every rule with positive net token savings costs at least 4.87 pp, and sophisticated event-triggered ("adaptive") probing is dominated by a simple window share.

Finding 3: Two taxes, one intrinsic. We separate the cost of stopping into an *accuracy tax*—you lose accuracy purely because you commit before reasoning finishes, independent of how you probe—and a *probe tax*—the token cost of the probes themselves, which turns the net savings of conservative rules negative under dense probing (Section 6). The accuracy tax is intrinsic and survives any probing scheme; the probe tax is a function of probe density. This distinction is what keeps the negative result robust: the 1.85 pp accuracy floor is not an artifact of dense probing.

Finding 4: The signal is the problem, not early exit. The failure is specific to *consensus*. Reproducing prior early-stoppers on the same frozen trajectories, a faithful Certaindex collapses by 56–70 pp (Section 7), but DEER (Yang et al., 2025)—which stops on the model's confidence in a trial answer at reasoning boundaries, *not* on probe agreement—stays within ~ 1 pp of full accuracy. Pushed into an online controller with a fast-commit path and a retained verification branch, this boundary-confidence signal saves $\sim 44\%$ of tokens at essentially neutral accuracy (+2.8 pp on one model; Section 8). Early exit is not hopeless; consensus is simply the wrong signal.

Contributions.

1. We define and quantify **false consensus** in reasoning models, including a calibration analysis showing window share systematically overstates reliability, and an error taxonomy of stopped-but-wrong cases (Section 3).

2. We introduce a **preregistered, probe-independent evaluation protocol** for early-exit rules—frozen trajectories, offline dense re-probing, a seven-dimensional rule schema, and fixed selection gates—and release the design and sweep artifacts (Section 4).

3. We report that, across 17,712 rules, **no stop in the searched space is simultaneously safe and token-saving** on the development set, and we characterize the full accuracy–savings frontier (Section 5).

4. We provide a **mechanism** (an accuracy-tax/probe-tax decomposition) and reproduce prior early-stoppers on identical frozen trajectories, showing the consensus-based ones collapse (Sections 6–7).

5. We give **constructive evidence** that a different signal (boundary confidence with a verification branch) escapes the accuracy tax, saving $\sim 44\%$ of tokens at neutral accuracy in an online deployment (Section 8).

2 Related Work

Probe-based early exit for reasoning. The methods we scrutinize periodically query a partial reasoning trajectory for its current answer and halt on agreement. Dynasor / Certaindex (Fu et al., 2024) formalize this with a "certainty index" over recent probe answers and use it to schedule early termination when serving reasoning programs; related work targets the *overthinking* of reasoning models (Chen et al., 2024) and proposes dynamic early exit (Yang et al., 2025). Our study is built on a fork of this line of work: we adopt the same probe mechanism (a short boxed-answer suffix) and the same notion of probe agreement, and ask how far *any* rule in this family can be pushed before it becomes unsafe.

Early-stopping self-consistency. The nearest neighbors to our setting stop *self-consistency* (Wang et al., 2023) early: Adaptive-Consistency (Aggarwal et al., 2023) and Early-Stopping Self-Consistency (Li et al., 2024) halt sampling once a majority among *completed* samples is statistically settled. Probe-based early exit is the online, *single-trajectory* analogue: it votes across snapshots of one in-progress trajectory rather than across finished samples. This difference is exactly why their stopping criteria do not transfer—snapshots of an unfinished trajectory are not exchangeable draws from a stable answer distribution, because the trajectory is still correcting itself (Section 3).

Self-correction. Our "recovery" statistic—most wrong intermediate answers are later overturned—is a form of intrinsic self-correction, but it differs from the prompted, post-hoc self-correction that Huang et al. (2024) find unreliable: we measure answer evolution *within* a single uninterrupted trajectory, not a second corrective pass. Our result is complementary: even when a model is not asked to self-correct, its mid-trajectory answer is not yet final.

Test-time scaling and verification. A large literature trades inference compute for accuracy via repeated sampling and search (Snell et al., 2024; Brown et al., 2024); early exit is the dual problem of *spending less* without losing accuracy. Process reward models and verifier-guided decoding (Lightman et al., 2024; Wang et al., 2024) score partial reasoning and are a natural source of a termination signal; we return to this in Section 9 as an alternative to consensus-based stopping.

Adaptive computation and token-level early exit. Adaptive computation time (Graves, 2016) and confident adaptive language modeling (Schuster et al., 2022) let models spend variable compute per input via a learned or thresholded halting signal at the *token or layer* level. Answer-level early exit for reasoning operates at the much coarser granularity of a whole answer and must contend with non-monotone answer trajectories; we quantify exactly this non-monotonicity.

Confidence and calibration. That model confidence overstates correctness is well documented for classifiers (Guo et al., 2017) and studied for LLMs via self-reported confidence (Kadavath et al., 2022) and semantic uncertainty (Kuhn et al., 2023). We report an analogous miscalibration for *probe consensus*: the share of recent probes agreeing on an answer is a poorly calibrated estimate of that answer's correctness, and is worst precisely in the regime a stopping rule would act on.

Preregistration. To keep an exhaustive rule search from silently overfitting to a favorable operating point, we fix the rule schema, splits, and acceptance gates before looking at held-out data (Nosek et al., 2025). To our knowledge this is among the first preregistered evaluations of early-exit rules for reasoning models; the protocol (Section 4) makes a *negative* result interpretable.

3 The False Consensus Phenomenon

We first establish, in an intervention-free setting, that intermediate probe consensus is not a reliable stopping signal.

3.1 Setup

We run DeepSeek-R1-Distill-Qwen-7B (vLLM, one A100-80G) on all 500 MATH500 problems (Lightman et al., 2024; Hendrycks et al., 2021) with a decode budget of 3072 tokens, temperature 0.6, top-p 0.95, and seed 42. Every 128 generated tokens (at most 24 probes per problem) we append the boxed-answer probe suffix and read off the model's current answer. The probe is a separate forward pass: the controller only *observes*, never halting or altering the main trajectory. This yields, per problem, the full answer trajectory a stopping rule would have access to. This exploratory study uses a deliberately tight 3072 budget so that many trajectories are truncated and the "continue vs. stop" contrast is visible within one A100; the preregistered sweep (Section 4) re-examines the phenomenon at the 16K/32K budgets used for the main results, and we report the recovery statistics there as well.

3.2 Agreement is not correctness

We compare two notions of probe agreement. *Cumulative* agreement (all non-empty probes so far concur) is nearly trustworthy: on the 87 problems that reach unanimous cumulative agreement, 98.9% are correct, with a single whole-trajectory false consensus. But cumulative agreement is not what an online stopping rule sees. The operative signal is *window* agreement—the last five probes concurring—and it is markedly less reliable: of the 338 problems reaching a unanimous last-five window, only 93.5% are correct, giving a 6.5% (22/338) false-consensus rate. Perfect *global* consensus is almost safe; the *local* consensus a controller must act on is not.

3.3 The cost lands at the stopping decision

We simulate a *naive consecutive-agreement* stop: halt at the first point where three consecutive probes agree, are non-empty, and are *certain* (the model emits the boxed answer with high token-level confidence; formal definition in Appendix A). This is a simplified heuristic, *not* the Certaindex algorithm, which we reproduce faithfully in Section 7. It fires on 416/500 problems. The committed answers are correct only 69.2% of the time, whereas letting those same problems run to completion (within the 3072 budget) reaches 85.6%—a 16.4 **pp accuracy loss** in exchange for an average saving of 1321 tokens, with 128 problems stopped on an outright wrong answer.

3.4 Recovery is common, and early stopping kills it

The loss is driven by *recovery*: continued reasoning overturns the intermediate answer far more often than intuition suggests. Of the 145 problems that at some point formed a three-probe consensus *different* from their final answer, 95 (65.5%) ended up correct; and of the 375 problems whose *first* probe was wrong, 76.3% eventually flipped to correct. Neither "first impression" nor "local consensus" is close to terminal.

Counterintuitively, *later* consensus is *less* trustworthy: consensus formed before 512 tokens is 87.4% correct, but consensus that only forms after 2048 tokens is 58.1% correct. Hard problems both converge slowly and converge to worse answers—exactly the problems on which an aggressive stopping rule does most damage.

3.5 Miscalibration

Window share is a poorly calibrated confidence estimate. Its calibration error—the gap between share and empirical accuracy, averaged over share bins (CCE)—is 0.080 (versus 0.149 for cumulative share), but the error is concentrated where a rule would act: at window share 0.8 the empirical accuracy is only 47.2% (n=36). A threshold tuned to "looks nearly unanimous" is operating precisely in the band where the signal is least reliable.

3.6 An error taxonomy

Hand-inspecting stopped-but-wrong cases (28 cases from the first 100 problems; AI-assisted initial labeling, [PENDING: full human review of the 134-case export]) yields *four* non-empty types out of five categories. By count: **(A)** numeric collapse—stable convergence to a wrong number—14/28; **(D)** derivation gaps—dropped roots/cases, unverified solutions—7/28; **(E)** format/option hallucination—a non-multiple-choice problem stably emitting a letter such as "B"—6/28, which is an artifact of the probe mechanism itself rather than a model belief; and **(B)** expression collapse—1/28. Type **(C)** (sign errors) did not occur in this sample. Type E is itself a methodological finding: roughly one in five "false consensuses" is a probe formatting artifact, implying that any deployed rule should filter answer *shape*.

Takeaway. These observations—local consensus unreliable, recovery common, late consensus worse, confidence miscalibrated—suggest that a good stopping rule must be much more than "agreement." The rest of the paper asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical. In the searched space, it cannot.

4 A Preregistered Test for Safe Early Exit

Section 3 refutes one heuristic. To ask whether *any* probe-based rule can succeed, we need (i) an evaluation that a rule cannot game by changing the reasoning it observes, and (ii) a selection procedure that cannot silently overfit to a lucky operating point. We preregister both.

4.1 Probe-independent, frozen trajectories

For each problem we generate exactly *one* main trajectory in a single request (caps: 16K tokens for MATH500 and AMC23, 32K for AIME24) and freeze it. All probing is then performed *offline* against fixed text prefixes of that frozen trajectory. Because probes never re-enter the decode, changing a probe schedule—or a stopping rule—cannot change the underlying reasoning; every rule is scored against the same trajectories. We build two offline probe banks: a *dense* bank that re-probes every 64 tokens with simple@32 (a 32-sample boxed-answer probe), and an *adaptive* bank that additionally fires event-triggered probes at entropy drops and at conclusion/reflection/answer markers. Token accounting charges every probe's output tokens to the rule that consumes it (Section 6).

4.2 A seven-dimensional rule schema

A stopping rule is a point in a schema with seven independent dimensions: **(1) probe schedule** (fixed interval, or adaptive event triggers); **(2) maturity** (minimum tokens or budget fraction before any stop is allowed, incl. an online-instability floor); **(3) evidence** (window share vs. latest-persistence vs. entropy-budget-fraction); **(4) persistence** (minimum consistent accepts and consensus span); **(5) certainty** (minimum fraction of supporting probes flagged certain); **(6) validity** (answer-shape filter, e.g., rejecting empty/letter answers); and **(7) history** (maximum answer switches, minimum stable span). Enumerating the preregistered grid over these dimensions yields 17,712 candidate rules spanning four families (`window_share`, `latest_persistence`, `entropy_budget_fraction`, `adaptive_event_probe`); formal definitions of each dimension are in Appendix A.

4.3 Models, benchmarks, splits

Development uses two models—DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B—on three benchmarks (MATH500, AMC23, AIME24), each at seeds 42/43/44: 18 model×benchmark×seed environments. Problems are partitioned *by problem id* into 60/20/20 train/dev/test splits (MATH500 300/100/100, AMC23 24/8/8, AIME24 18/6/6), so no problem leaks across splits. Confirmation reserves seeds 45/46/47 plus two unseen models—DeepSeek-R1-Distill-Llama-8B and Distill-Qwen-32B—all evaluated on the *test* split only.

4.4 Preregistered operating points and gates

We fix three operating points and their acceptance gates *before* looking at held-out data (Table 1). Selection maximizes worst-case token savings subject to worst-case accuracy-drop caps and a positive-savings-fraction (psf) floor. A rule is accepted at an operating point only if it satisfies that point's gate on train+dev.

Operating point	Max model drop	Max bench drop	psf floor
conservative	1.5 pp	2.0 pp	0.80
balanced	2.5 pp	3.0 pp	0.80
token_efficient	4.0 pp	5.0 pp	0.70

Table 1: Preregistered acceptance gates. "Max model/bench drop" cap the worst per-model and per-benchmark accuracy loss; psf is the fraction of environments with strictly positive net savings. Gates are fixed in advance and *not* relaxed post hoc.

4.5 Metric resolution

The per-*model* drop pools all of a model's dev instances (≈ 114 problems per seed, 342 over three seeds), so its resolution is fine (∼ 0.3 pp) and the 1.5 pp conservative cap is meaningful. The per-*benchmark* drop is coarser on the small competition sets: one AIME24 problem is 16.7 pp of a 6-problem dev split (5.6 pp pooled over three seeds) and one AMC23 problem is 12.5 pp (4.2 pp pooled), so a 2.0 pp per-benchmark cap is finer than a single problem can resolve there. We therefore treat the **per-model gate as primary** and the per-benchmark gate as a *diagnostic*, and we report the accuracy floor of Section 5 as a per-model (pooled) quantity that is not driven by single-problem quantization. Confidence intervals for all headline figures are [PENDING: bootstrap over problems and seeds].

4.6 Preregistration commitments

Three commitments make the result interpretable. **(C1)** The test split and both confirmation models are never read during rule selection. **(C2)** The conservative gate in Table 1 is not relaxed after seeing results; if no rule passes, that is reported as the finding rather than engineered away. The balanced and token_efficient points are reported only as clearly-labeled post-hoc analysis, never as a selected operating point. **(C3)** The frozen rule set is fixed on train+dev and its hash recorded before any test/confirmation evaluation. Section 5 reports what happened when we applied C2.

5 Results: No Safe-and-Saving Stop in the Searched Space

We aggregate the sweep into per-environment metrics—637,632 rows = 17,712 rules × 18 environments × 2 splits (train, dev)— and compute, for each rule, its worst per-model and per-benchmark accuracy drop and its net token savings (Section 6 defines the accounting). All numbers below are on the development set; held-out confirmation is [PENDING: deferred to C3]. Applying commitment C2 to the preregistered conservative gate yields the paper's central result.

5.1 The conservative gate is empty

No rule in the 17,712**-candidate space clears the conservative gate** (per-model drop ≤ 1.5 pp, per-benchmark ≤ 2.0 pp, psf ≥ 0.8). The Pareto frontier over (worst-case drop, net savings) contains 93 rules, and none is admissible. The obstruction is not a narrow miss:

- the *smallest* per-model accuracy drop anywhere in the space is 1.85 pp (a pooled per-model quantity, not a single-problem artifact; Section 4.5)—already above the 1.5 pp gate—and the rules achieving it have *negative* net savings (20th-percentile saving $\approx -11\%$; only 2.8% of environments save at all);
- *no* rule achieves a near-zero drop: over all 17,712 candidate rules the per-model drop has percentiles p_1=3.37, p_5=4.26, p_{25}=10.7, median 20.1 pp;
- obtaining even three distinct rules with *positive* net savings requires accepting a per-model drop of at least 4.87 pp—and this holds whether the psf floor is set to 0.8, 0.7, 0.6, or 0.5.

Direction of effect. The failure is systematic, not sampling noise. *Every* one of the 17,712 candidate rules incurs a positive worst-case per-model accuracy drop (minimum 1.85 pp, median 20.1 pp); *zero* rules reach break-even. Across all 637,632 rule–environment evaluations, accuracy strictly *falls* in 67.7% and strictly *rises* in only 6.7% (unchanged in the remaining 25.5%), for a mean drop of 10.9 pp. The rare gains are small and do not concentrate in any rule or family. This is exactly the direction our mechanism (Section 6) predicts: consensus forms while the answer is still moving, so acting on it removes correct recoveries more often than it banks a settled trajectory.

Two claims should be kept separate. The **safety** result—no rule reaches even 1.5 pp, the floor being 1.85 pp—is a statement about accuracy and, as Section 6 argues, is independent of how densely we probe. Only the per-model condition is binding here; the per-benchmark cap is a diagnostic (Section 4.5). This particular margin is thin: a bootstrap over the nine benchmark×seed environments puts the least-bad rule's worst-case per-model drop at 1.85 pp with a 95% CI of $[0.0, 5.6]$ pp, which includes break-even, so we do *not* rest the result on that single rule's point estimate. We rest it instead on the two facts that do not depend on it: all 17,712 rules lose accuracy (above), and even this least-bad rule spends *extra* tokens (its net savings are -8 to -9%). Held-out confirmation (C3) then tests whether the floor reproduces out of distribution. The **savings** result—positive net savings costs ≥ 4.87 pp—additionally depends on our dense-probe accounting and is therefore the more caveated of the two. Within the preregistered space one must choose between a stop that is (nearly) safe but costs tokens and a stop that saves tokens but is unsafe.

Rule family	Per-model drop (pp)	Net savings at that point
`entropy_budget_fraction`	1.85	negative ($q_{20}\approx-11\%$)
`window_share`	4.87	≈ 0 ($q_{20}\approx0.3\%$)
`latest_persistence`	8.20	$q_{20}\approx9.7\%$
`adaptive_event_probe`	9.70	$q_{20}\approx4.5\%$

Table 2: Best operating point per rule family on dev. For `entropy_budget_fraction` we show its *absolute* minimum drop (which has no positive-savings point); for the other three families we show the minimum worst-case per-model drop *among positive-savings points*. The safest family cannot save; the most complex family (adaptive) is dominated by the simplest (window share).

5.2 Family frontier

Table 2 gives each family's best attainable operating point. The ordering is informative. The simplest evidence signal (`window_share`) gives the best safe-with-positive-savings point (4.87 pp); `entropy_budget_fraction` can push the drop down to 1.85 pp but only by stopping so rarely that savings go negative; and `adaptive_event_probe`—the most sophisticated family, with entropy- and marker-triggered probing—is dominated, needing 9.70 pp for a positive-savings point.

5.3 Adaptive probing does not help

Because event-triggered probing is the natural "smarter" fix, we isolate its frontier (Table 3). Even its most conservative point loses 5.88 pp on average (worst environment -9.70 pp) for a 14.2% mean saving; becoming more aggressive trades linearly into larger losses (-15.03 pp macro for a 37.3% saving). Entropy-drop and conclusion-marker trig-

Adaptive point	Trigger	Mean save	q_{20} save	Macro Δacc
conservative	entropy drop	14.2%	4.5%	-5.88 pp
moderate	conclusion	25.9%	20.0%	-9.11 pp
aggressive	conclusion	37.3%	27.1%	-15.03 pp

Table 3: The `adaptive_event_probe` frontier on dev. Worst per-model/per-benchmark drops for the three points are $-9.70/-10.42$, $-13.65/-12.50$, and $-19.28/-18.75$ pp respectively. Reproduced exactly from the released sweep archive.

gers buy savings only by paying accuracy—at its best positive-savings point the adaptive family is dominated by simple window consensus. Sophistication in *where* to probe cannot fix the fact that the answer is still moving.

5.4 Main comparison

Table 4 places our best rule-based operating points against reference methods on the 18 dev environments. "Ours" denotes the best rule in our schema *targeting* each operating point; none passes the conservative gate (Δacc is macro-averaged, whereas the 1.85 pp floor above is the worst-case per-model drop for the same rule). The full comparison against prior-work early-stoppers is completed in Section 7.

5.5 Held-out confirmation

Per commitment C3, confirmation on the test split and on the two unseen models (Llama-8B, Qwen-32B) is [PENDING: run only after a frozen rule set / operating point is fixed; see Section 9]. Because the development result is that *no* rule clears the conservative gate, confirmation is framed as validating the frontier lower bound rather than a selected rule.

6 Mechanism: Two Taxes, One Intrinsic

Why is the safe-and-saving corner empty? We separate two distinct costs of stopping and argue that only one of them depends on how we probe.

6.1 Accounting

For a rule that stops a problem at token position s after consuming probe output p, we charge total decode tokens $T = s + p$ and compare against the frozen baseline B (the full trajectory, capped at budget). Net savings is $(B - T)/B$; the accuracy drop compares the correctness against ground truth of the committed answer at s with that of the frozen final answer (it is not an agreement rate between the two). Two quantities therefore move independently: *where* the rule stops (which sets both the accuracy drop and the main-token saving s) and *how much it probes* to get there (which adds p).

6.2 The accuracy tax is intrinsic

The accuracy drop depends only on the stop position s, not on p: committing to the answer at s forgoes whatever correction the trajectory would have made after s. This is the quantity that never reaches zero—the 1.85 pp floor of Section 5 is an *accuracy tax* that no probing scheme can remove, because it is a property of the reasoning trajectory, not of the probe. A *direction-of-effect* test operationalizes this: among stops where the committed answer differs in correctness from the final answer, we measure the ratio of (final-correct, stop-wrong) to (final-wrong, stop-correct). Pure sampling noise predicts $\approx$ 1:1; a genuine "continued reasoning corrects" effect predicts a ratio $\gg 1$. On the frozen replay this ratio is **35**:**1** for the naive consensus stopper (final-correct/stop-wrong $= 1055$ vs. final-wrong/stop-correct $= 30$; 34.3:1 and 35.9:1 on DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B separately), and 15–18:1 for the more conservative variants—all far above the 1:1 of sampling noise. Stopping on consensus therefore destroys a correct-in-the-end answer roughly 35 times more often than it rescues a wrong one, corroborating the intervention-study recovery statistics of Section 3 (65.5%, 76.3%). The accuracy tax is a real, directional effect, not sampling noise.

6.3 The probe tax is a design choice

The second cost is the probe output p. Under dense re-probing (every 64 tokens, 32-token probes), a rule that stops late pays p over a long prefix, which is what turns the *net* savings of the safe (entropy) family negative even though its *gross* savings $(B - s)/B$ is positive. Unlike the accuracy tax, the probe tax scales with probe density and is reducible (sparser probes, shorter probes, or KV-cache reuse). We therefore report savings two ways:

6.4 Why the distinction matters

Separating the taxes pre-empts the natural objection that our negative result is an artifact of dense probing. It is not: the *savings collapse* is partly a probe-tax artifact and is addressable, but the *accuracy floor*—the finding that even the safest stop costs ≥ 1.85 pp—is probe-independent

Method	Accuracy	Δacc (macro)	Net token saving
Full generation	TBD	—	0%
Fixed budget	TBD	TBD	TBD
Ours (target conservative)	TBD	≈ -0.9 pp	$\approx -4\%$
Ours (target balanced)	TBD	TBD	TBD
Ours (target token_efficient)	TBD	≈ -5.6 pp	$\approx 19.6\%$

Table 4: Our best swept operating points on dev (macro over 18 environments). "Ours" rows are the best rule in our schema targeting each operating point; the target-conservative rule is roughly accuracy-neutral in macro but spends $\sim$4% *extra* tokens because of the probe tax (Section 6) and does not pass the per-model gate. Prior-work early-stoppers on the same trajectories are reported in Table 6 (Section 7). Accuracy cells marked TBD are being finalized from the sweep aggregation.

Operating point	Gross saving $(B-s)/B$	Net saving $(B-s-p)/B$
Ours (target conservative)	TBD	$\approx -4\%$
Ours (target balanced)	TBD	TBD
Ours (target token_efficient)	TBD	$\approx 19.6\%$

Table 5: Gross vs. net savings. The gap is the probe tax. The target-conservative point's negative net saving is a probing-density artifact, not evidence that stopping wastes tokens; the accuracy tax above it is not.

and survives any probing scheme. This probe-independence rests on one premise: the sweep's probe grid (intervals 64/128/256 tokens) makes essentially every stop position reachable, and the 1.85 pp floor is the minimum over these schedules; a sparser schedule can only stop later or at a subset of positions, never at a safer one that the grid missed. The two families of prior work we test next fail on the accuracy axis, not the probe axis.

7 Prior Early-Stoppers on the Same Trajectories

Our sweep searches *consensus*-based rules. To place prior work on the same footing—and to test whether the failure is specific to the consensus signal—we reproduce three published early-stoppers in a common harness: the same frozen trajectories, the same simple@32 probe bank, and the same token accounting (probe output tokens included). All numbers are dev macro-averages over three benchmarks per model (18 environments); they are reproductions on our frozen trajectories, not the original papers' end-to-end results.

Methods. **CertaIndex** (Fu et al., 2024): the faithful Certaindex share-threshold stop—the direct descendant of the consensus family we sweep. **DEER** (Yang et al., 2025): stops at reasoning-boundary ("`Wait`") tokens using the model's *own* confidence in a short trial answer—a different signal from probe agreement. **TJE** (Anonymous, 2026): a token-level early-exit stop.

The consensus signal is the problem, not early exit. Table 6 and Figure 1 sharpen the negative result. Every method that stops on *probe agreement*—the swept schema and CertaIndex alike—either fails the safety bar or (CertaIndex) collapses outright. The lone method that stays near full accuracy, DEER, does not use probe agreement at all: it stops on the model's confidence in a trial answer emitted at a reasoning boundary. This is the first direct evidence that the failure we characterize is a property of the *consensus signal*, and it motivates asking how far a boundary-confidence signal can be pushed (Section 8). We caution that DEER's advantage is model-dependent (+0.8 pp on Qwen3-8B but −4.8 pp on DeepSeek-R1-Distill-Qwen-7B) and that these are frozen-trajectory reproductions.

8 A Signal That Works: Boundary-Confidence Early Exit

Section 7 showed that the one prior method staying near full accuracy, DEER, reads a different signal. We now push that signal into a deployment-style online controller and find it recovers substantial savings at neutral accuracy—the constructive counterpart to our negative result. These results are exploratory (single seed, outside the preregistered sweep); we report them to establish that the failure is about the *signal*, not early exit.

8.1 From trial-answer confidence to a verification branch

DEER emits a short trial answer at reasoning-boundary ("`Wait`") tokens and stops once the model's confidence in that trial answer (the mean probability of its answer tokens) exceeds a threshold. On our development trajectories the mecha-

Method	Model	Accuracy	Δacc	Fair token saving	Stop rate	Signal
Full generation	Qwen3-8B	85.4%	—	0%	0%	—
Full generation	DeepSeek-R1-Distill-Qwen-7B	79.8%	—	0%	0%	—
CertaIndex	Qwen3-8B	15.3%	−70.1 pp	90.1%	99.8%	consensus
CertaIndex	DeepSeek-R1-Distill-Qwen-7B	23.9%	−55.9 pp	76.7%	98.7%	consensus
TJE	Qwen3-8B	85.0%	−0.4 pp	2.0%	22.5%	token-level
TJE	DeepSeek-R1-Distill-Qwen-7B	60.7%	−19.1 pp	65.0%	93.4%	token-level
DEER	Qwen3-8B	86.2%	**+0.8** pp	16.3%	41.4%	boundary conf.
DEER	DeepSeek-R1-Distill-Qwen-7B	74.9%	−4.8 pp	20.2%	56.1%	boundary conf.

Table 6: Prior early-stoppers reproduced on our frozen trajectories (dev, macro over three benchmarks; probe tokens charged). CertaIndex—the consensus descendant—**collapses** (−56 to −70 pp) while saving 77–90% of tokens: it fires on almost every problem, very early, exactly the false-consensus trap of Section 3. TJE is unsafe on the weaker model. **DEER, which reads a different signal** (boundary confidence, not probe agreement), is the only method that stays near full accuracy—+0.8 pp on Qwen3-8B—while still saving tokens.

Method	Model	Δacc	Fair saving
Inspired	Qwen3-8B	+2.8 pp	45.6%
Inspired	DeepSeek-7B	−3.5 pp	41.8%
Inspired	macro	−0.4 pp	43.7%
DEER ref.	macro	−1.3 pp	28.3%

Table 7: Online boundary-confidence early exit (dev, seed 42, 456 trajectories). The inspired fast-path + verification-branch controller keeps macro accuracy within 0.4 pp of full generation while saving $\sim 44\%$ of tokens, and dominates a faithful online DEER reference on savings at equal accuracy. Exploratory: single seed, small AMC23/AIME24 sets.

nism has a flaw: DEER measures confidence on the trial answer but then *delivers a separately generated readout*, and the two disagree on 14.6% of triggered problems. The extra readout averages 470 tokens yet yields no net accuracy gain (trial 88.9% vs. readout 88.5%), though it corrects some cases on some models. We therefore evaluate an *inspired* controller that (i) *fast-commits* the trial answer when confidence is very high (> 0.995), skipping the redundant readout, and (ii) otherwise opens a *retained verification branch* that commits only if a second high-confidence answer is mathematically equivalent, and otherwise keeps reasoning. Both run in a single online engine that generates the main trajectory live and probes only at scheduled boundaries—no frozen suffixes.

8.2 Online deployment results

Table 7 reports an online run over the 18-environment development set (seed 42; 456 trajectories). The inspired controller holds macro accuracy essentially flat (−0.36 pp; +2.78 pp on Qwen3-8B) while saving $\sim 44\%$ of tokens, with 69% of problems fast-committed and 15% routed through a verification branch. Against a faithful online DEER reference, a stratified paired bootstrap (10,000 resamples) gives an accuracy difference of +0.97 pp (95% CI $[-10.5, +12.6]$) and a token-saving advantage of +15.3% (95% CI $[+7.3, +22.9]$): the controller saves significantly more at indistinguishable accuracy.

8.3 Relation to the preregistration

This controller is *not* part of the preregistered consensus sweep: it uses a different signal, a separate configuration, and its own results directory, and it would require its own train/dev→test validation before any confirmed claim. It does not modify the sweep's candidates, gates, or selected rules (there were none). We therefore present it as a positive *signal*, not a finished method: it shows that stepping outside the consensus-signal space escapes the accuracy tax that Section 5 proved unavoidable inside it.

9 Discussion

What the negative result does and does not say. It says: within a large, preregistered space of *consensus-based* stopping rules, and under a dense-probe measurement regime, no rule on the development set is both safe (per-model drop ≤ 1.5 pp) and net token-saving. It does *not* say early exit is impossible in principle, nor that it is impossible outside the searched schema. The result localizes the failure: the signal these rules consume—agreement among probes of an unfinished trajectory—is intrinsically non-terminal, so no threshold on that signal escapes the accuracy tax.

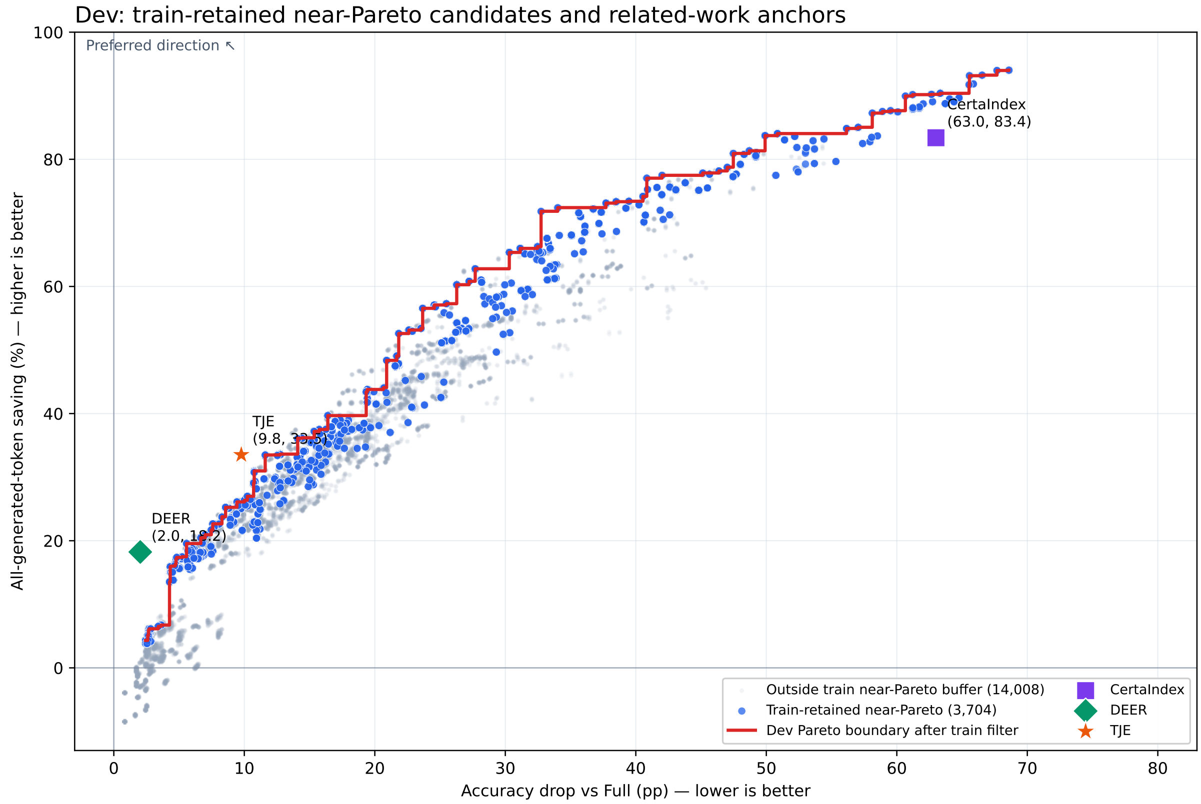

Figure 1: Every swept consensus rule (dev) against the three prior early-stoppers. Each point is one of the 17,712 candidate rules; blue are the 3,704 surviving a near-Pareto train filter, the red step is the dev Pareto boundary, and the markers are the reproduced anchors. The safe-and-saving corner (top-left: small drop *and* large saving) is empty —the boundary only climbs as the accuracy drop grows. **CertaIndex** sits at the far right (63 pp drop), the consensus family's collapse; **TJE** at 9.8 pp; and **DEER**—a boundary-confidence signal *outside* the swept space—occupies the low-drop corner (2.0 pp, 18%), above the consensus Pareto boundary at the same accuracy drop. The saving axis is all-generated tokens (the related-work convention of Table 6); net probe-charged saving (Section 6) is lower and can be negative.

Implication: change the signal, not the threshold. Much prior effort has gone into *tuning* consensus-based stops (Fu et al., 2024; Aggarwal et al., 2023; Li et al., 2024). Our frontier shows the ceiling of that approach is low, and Section 8 gives direct evidence that a different termination signal—one about whether the *trajectory itself* will still change—does better: boundary confidence with a verification branch saves $\sim 44\%$ of tokens at neutral accuracy. Other candidates point the same way: verifier- or process-reward-based termination (Lightman et al., 2024; Wang et al., 2024) and learned halting in the spirit of adaptive computation (Graves, 2016; Schuster et al., 2022). The lesson is not that early exit is hopeless but that it should read the model's own forward-looking confidence, not the agreement of backward-looking probes.

Two live extensions. Two follow-ups are compatible with the preregistration and would strengthen the paper without touching the test split. **(A)** *Confirmation as lower-bound validation*: evaluate the family-best points on the held-out test split and on the two unseen models, showing the ≥ 4.87 pp price of savings reproduces out of distribution. **(B)** *Attacking the probe tax*: a newly preregistered extension with sparser probes / KV-cache reuse could recover positive *net* savings at the safe end, converting "no safe-and-saving stop in the searched space" into "…exists only if probing is made this much cheaper." **(C)** *Validating boundary confidence*: the online controller of Section 8 is single-seed and model-dependent ($+2.8$ pp on Qwen3-8B but -3.5 pp on DeepSeek-R1-Distill-Qwen-7B); a multi-seed train/dev$\rightarrow$test evaluation would turn the positive signal into a confirmed method. All three are additive to the frozen findings of Sections 3–6.

10 Conclusion

We asked whether probe-based early exit for reasoning LLMs can be made safe by choosing a better stopping rule. For the consensus-based family the answer is no: intermediate probe consensus is systematically non-terminal (a unanimous window is correct only 93.5% of the time; a naive stop loses 16.4 pp), a preregistered search over 17,712 rules finds none that is both safe and token-saving on the development set (the safest costs 1.85 pp), and a faithful Certaindex reproduction collapses by 56–70 pp. But the failure is specific to the signal, not to early exit: a boundary-confidence stop with a verification branch—which never reads probe agreement—stays near full accuracy and, in an online deployment, saves $\sim$44% of tokens at essentially neutral accuracy. Separating the intrinsic accuracy tax from the reducible probe tax explains why consensus stalls. The practical takeaway is that safe early exit for reasoning should read the model's own forward-looking confidence at reasoning boundaries, not the agreement of backward-looking probes—and confirming the boundary-confidence signal across seeds and held-out models is the immediate next step.

Limitations

Development-set scope. The central result—no rule clears the conservative gate—is established on the 18 development environments (two models, three benchmarks, seeds 42/43/44) using the train and dev splits. Held-out confirmation on the test split and on the two unseen models (Llama-8B, Qwen-32B) is [PENDING: not yet run]; the claim is therefore currently a development-set finding. Because the effect sizes are large (median frontier drop 20.1 pp), we expect confirmation to reproduce the direction, but the out-of-distribution magnitudes are not yet measured.

Small held-out sets on hard benchmarks. AIME24 and AMC23 are small; the dev split has only 6 and 8 problems per seed, so a single problem is a large fraction. A worst-case per-model drop on AIME24 can correspond to two or three problems, and is statistically noisy. We mitigate by aggregating over seeds and reporting macro averages, but individual per-benchmark cells should be read with this in mind.

Probe-tax dependence of the savings axis. Our net-savings numbers charge dense re-probing (every 64 tokens, 32-token probes). A different probe schedule—sparser, shorter, or with KV-cache reuse—would change the *savings* axis and could move some rules from negative to positive net savings. We separate gross from net savings (Section 6) precisely so that the probe-tax-dependent and probe-tax-independent parts of the result are distinguishable; the accuracy-tax result does not depend on the probe schedule.

Single probe suffix and rule space. We use one boxed-answer probe suffix (simple@32) and one preregistered rule schema. Other probe wordings (e.g., the original Certaindex phrasing) and rule dimensions outside this schema are not searched; a negative result is a statement about the space searched, which we report explicitly (17,712 rules, four families, seven dimensions).

Boundary-confidence results are exploratory. The positive result of Section 8—boundary confidence with a verification branch—is a single-seed (seed 42) online run, is not part of the preregistered sweep, and is model-dependent (+2.8 pp on Qwen3-8B but −3.5 pp on DeepSeek-R1-Distill-Qwen-7B); its online multi-request sampling path also differs from the single-request full baseline, making the vs-full comparison approximate. It has an invalid-trial rate of 21.7% and no test-split confirmation. It should be read as evidence that a non-consensus signal escapes the accuracy tax, not as a finished, validated method.

Domain. All benchmarks are competition mathematics with checkable final answers. Whether false consensus and the accuracy tax transfer to open-ended reasoning, code, or agentic tasks is out of scope here.

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A Rule Schema Details

The seven-dimensional schema of Section 4 enumerates to 17,712 candidate rules. Each rule fixes:

1. **Probe schedule**: fixed interval $\in \{64, 128, 256\}$ tokens, or `adaptive_event` with entropy-drop and marker triggers and a fallback interval.

2. **Maturity**: `fixed_tokens` (minimum tokens), `budget_fraction` (minimum fraction of the decode budget), or `online_instability` (a floor plus a same-answer stability check).

3. **Evidence**: window share, latest persistence, or entropy-budget-fraction candidate selection.

4. **Persistence**: minimum consistent accepts and minimum consensus span in tokens.

5. **Certainty**: minimum fraction of supporting probes flagged certain.

6. **Validity**: answer-shape filter (reject empty / single-letter answers to suppress the Type-E probe artifact of Section 3).

7. **History**: maximum answer switches within a window and minimum stable span since the last switch.

Certainty. A probe is *certain* when its boxed answer is emitted with high token-level confidence: concretely, the mean probability of the answer tokens under the probe forward pass exceeds a fixed threshold (the Certaindex certainty flag). For the simple@32 bank this is aggregated over the 32 samples as the fraction agreeing on the modal answer. Dimension 5 gates a stop on the fraction of the supporting probes that are certain.

Entropy triggers. The `adaptive_event` schedule fires an extra probe when the token-level predictive entropy of the main decode drops by at least ΔH relative to a trailing baseline (an "the model just committed" signal), and at conclusion/reflection/answer marker tokens (e.g., "therefore", "the answer is"), subject to a minimum inter-probe gap and a fallback interval.

B Preregistration Text

The protocol version, split manifest hash, candidate-rule generation script, and the three operating-point gates (Table 1) were fixed before any held-out evaluation. Commitments C1–C3 (Section 4) were recorded with the protocol. When the conservative gate proved empty, we followed C2 and report the empty gate as the finding rather than relaxing it; the balanced/token_efficient points are reported only as clearly-labeled post-hoc analysis and would be re-preregistered if used for selection.

C Reproducibility

Frozen trajectories, offline probe banks, the sweep archive (637,632 metric rows), and the aggregation scripts are released. Table 3 was reproduced to the reported precision directly from the released sweep archive. Grading uses a robust answer-equivalence check that tries the raw and normalized reference forms and both string and math-expression equality; per-problem correctness for the frozen baseline is recomputed at replay time rather than trusted from collection. The grader's measured error rate against a hand-checked sample is [PENDING: report grader error rate].

D Additional Tables and Figures

[PENDING: frontier scatter (worst-case drop vs. net savings, families colored); window-share reliability diagram; per-model $\times$ benchmark full-accuracy table; direction-of-effect counts; gross/net decomposition by family; baseline sweeps.]